

Text 1

Nearly 2,000 years ago, as the Romans began to pull out of Scotland, they left behind a curious treasure: 10 tons of nails, nearly a million of the things. The nail hoard was discovered in 1960 in a four-metre-deep pit covered by two metres of gravel.

Why had the Romans buried a million nails? The likely explanation is that the withdrawal was rushed, and they didn't want the local Caledonians getting their hands on 10 tons of weapon-grade iron. The Romans buried the nails so deep that they would not be discovered for almost two millennia.

Later civilisations would value the skilled blacksmith's labour in a nail even more than the raw material. As Roma Agrawal explains in her new delightful book *Nuts and Bolts*, early 17th-century Virginians would sometimes burn down their homes if they were planning to relocate. This was an attempt to recover the valuable nails, which could be reused after sifting the ashes. The idea that one might burn down an entire house just to reclaim the nails underlines how scarce, costly and valuable the simple-seeming technology was.

The price of nails fell by 90% between the late 1700s and mid-1900s, as economist Daniel Sichel points out in a research paper. According to Sichel, although the falling price of nails was driven partly by cheaper iron and cheaper energy, most of the credit goes to nail manufacturers who simply found more efficient ways to turn steel into nails.

Nails themselves have changed over the years, but Sichel studied them because they haven't changed much. Roman lamps and Roman chariots are very different from LED strips and sports cars, but Roman nails are still clearly nails. It would be absurd to try to track the changing price of sports cars since 1695, but to ask the same question of nails makes perfect sense.

I make no apology for being obsessed by a particular feature of everyday objects: their price. I am an economist, after all. After writing two books about the history of inventions, one thing I've learnt is that while it is the enchantingly sophisticated technologies that get all the hype, it's the cheap technologies that change the world.

The Gutenberg printing press transformed civilisation not by changing the nature of writing but by changing its cost—and it would have achieved little without a parallel collapse in the price of surfaces to write on, thanks to an often-overlooked technology called paper. Solar panels had a few niche uses until they became cheap; now they are transforming the global energy system.